Avicenna Arya Chandramukti (展沐迪)

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Final-year double-degree Electrical Engineering undergraduate at National Sun Yat-sen University (NSYSU, Taiwan) and Institut Teknologi Sepuluh Nopember (ITS, Indonesia), maintaining a 3.76 GPA. Multidisciplinary technical background spanning IC packaging, SMT assembly, multi-layer KiCad PCB design, embedded C/C++ firmware, FPGA Verilog logic synthesis, and power management systems. Demonstrated success leading hardware teams in international robotics competitions and managing campus-wide student governance. Seeking graduate engineering roles in Taiwan's semiconductor, hardware, and electronics sectors.

EDUCATION

National Sun Yat-Sen University - Kaohsiung, Taiwan
Bachelor of Science in Electrical Engineering (Double Degree Program) | GPA: 3.76 / 4.30
Expected Graduation: July 2027

Sepuluh Nopember Institute of Technology – Surabaya, Indonesia
Bachelor of Science in Electrical Engineering | GPA: 3.54 / 4.00
Expected Graduation: July 2027

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ENGINEERING & INDUSTRY EXPERIENCE

TFME (Teaching Factory Manufacturing of Electronics) - Batam, Indonesia
IC Packaging & SMT Assembly Intern | July 2026 – August 2026

- Operated and monitored Surface Mount Technology (SMT) pick-and-place machinery, wire bonders, die attach units, and reflow soldering ovens for microelectronics manufacturing.
- Executed structural quality control, failure inspection, and RF signal testing on assembled IC packages and microelectronic modules.
- Authored comprehensive technical reports, workflow documentation, and standard operating procedures (SOPs) for semiconductor packaging and assembly lines.

6th Global Project-Based Learning (gPBL) Competition - KMITL, Bangkok
Robotics Hardware & Firmware Lead | September 2026

- Awarded "Best Presentation Award" for designing and building an autonomous time-trial line-following robot.
- Engineered ESP32 microcontroller hardware utilizing ESP-NOW protocol for low-latency peer-to-peer wireless telemetry.
- Hardcoded embedded C++ motor control algorithms, PID feedback loops, and sensor fusion routines for real-time execution.

ORGANIZATIONS

Student Executive Board (BEM ITS) - Surabaya, Indonesia
Student Representative & Event Coordinator | December 2024 – September 2025

- Coordinated cross-departmental student advocacy and organized campus-wide engineering seminars for 1,200+ students.
- Managed logistics, speaker invitations, and budgeting for academic workshops and faculty events.

TECHNICAL PROJECTS

8-Series Intelligent Battery Management System (BMS) | Lead Hardware & Firmware Developer (Feb 2026 – Present)

- Designed a custom multi-layer BMS PCB layout in KiCad integrated with LTC6803G-4 IC for voltage/thermal monitoring and active cell balancing.
- Programmed C++ firmware on ESP32/STM32 microcontrollers to communicate via SPI and transmit real-time telemetry data.
- Built a C# (.NET WinForms) desktop telemetry dashboard connected with Firebase for live battery parameter visualization and data logging.

FPGA Digital Logic Design & Verification | DE1-SoC Hardware Project (2025)

- Implemented Verilog HDL modules in Quartus Prime for DE1-SoC FPGA, featuring UART communication interfaces and PWM motor controllers.
- Conducted hardware logic verification, signal timing analysis, and waveform debugging using oscilloscopes and logic analyzers.

TECHNICAL SKILLS

- Hardware Design & PCB: KiCad PCB Layout, LTC6803G-4 BMS IC, Schematic Capture, Gerber File Generation, LTspice, PSIM.
- Semiconductor Manufacturing: IC Packaging, Surface Mount Technology (SMT), Wire Bonding, Die Attach, Reflow Ovens, RF Testing, Failure Analysis.
- Microcontrollers & FPGAs: ESP32, STM32, Arduino, DE1-SoC FPGA, C/C++, Verilog HDL, MicroPython.
- Testing & Instrumentation: Digital Oscilloscopes, Logic Analyzers, Signal Generators, Digital Multimeters, Function Generators.
- Software & Engineering Tools: Quartus Prime, MATLAB/Simulink, VS Code, Visual Studio (C# .NET), Proteus, ANSYS, HTML/CSS/JavaScript.

CERTIFICATION

- **Awards**: Best Presentation Award (6th gPBL Competition 2026)
- **Certifications**: Programmable Logic Controller (PLC) Certification (UGM), MATLAB Certificate of Completion (ITS)
- **Languages**: English (Professional Proficiency, TOEFL ITP 573), Mandarin Chinese (Basic / Elementary), Indonesian (Native)